

Ethan Mahajan

Mechanical Engineering Student

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📍 Dacula, GA

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Professional Summary

- Mechanical engineering student with experience in CAD design, additive manufacturing, electromechanical prototyping, and freelance engineering consulting. Completed 20+ client projects while independently designing and building a high-speed CoreXY 3D printer.

Education

Georgia Institute of Technology, Mechanical Engineering

Atlanta, GA

- GPA: 4.00

Aug 2026 – May 2029

- Relevant Coursework: Principles & Applications of Engineering Materials; Engineering Graphics

Mill Creek High School, STEM

Hoschton, GA

- Weighted GPA: 4.63; SAT: 1540 (Top 1% Nationally)
- AP Scholar with Distinction; National Honor Society
- Physics and Math Scholar Award (1 of 700 Students)
- Relevant Coursework: AP Physics; AP Calculus; AP Chemistry

Aug 2022 – May 2026

Skills

CAD & Design: Fusion 360, Solidworks, Onshape, FreeCAD, Technical Drawings, Design for Manufacturing

Manufacturing & Fabrication: FDM 3D Printing, Rapid Prototyping, Mechanical Assembly, Woodworking, Metalworking, Basic Metrology

Electronics & Hardware: PCB Design, Soldering, Wiring, Crimping, Multimeter Diagnostics, CAN Bus, Connector Assembly

Machinery & Tools: FDM 3D Printers, Laser Engravers, Manual & CNC Milling, & Wood Saws

Software & Systems: Microsoft Office, Google Workspace, Github, Windows, Linux, MacOS, Klipper, Java, Python, & Markdown

Communication & Project Work: Technical & Client Communication, Team Collaboration, Project Management & Documentation, Online Tutoring, Lesson Planning, Time Management

Projects

FDM 3D Printer Design, Prototyping, & Revising

June 2024 – July 2026

Created a high-speed CoreXY 3D printer by overhauling an older machine.

- Designed a frame and motion system in CAD to meet mechanical and thermal specifications.
- Verified component tolerances and clearances before fabrication.
- Fabricated a prototype using metalworking machines and additive manufacturing.
- Configured Kalico Klipper firmware for high-speed, quality 3D prints.
- Installed CAN-bus toolhead electronics to compartmentalize for easier maintenance.
- Used vibration analysis and input shaping to improve rigidity, reliability, and print speed.

Air Purifier

July 2026 – present

Smart DIY air purifier that reuses a 120 mm computer case fan and responds to room particulate-matter levels.

- Designed a control concept for automatic fan-speed adjustment and filter replacement detection.
- Completed electrical schematics in KiCad; PCB placement and testing are coming soon.

Laser-Engraved Chess Board

Aug 2026 – present

In-progress Invention Studio project combining CAD, laser engraving, and planned CNC-machined metal pieces.

- Developed a quickly manufacturable board design and planned low-center-of-mass pieces for stability.
- Completed approximately half of the board fabrication; CNC-machined metal pieces are coming soon.

Experience

Fiverr, Freelance CAD Designer & Manufacturing Consultant

Remote

Completed 20+ custom engineering contracts involving CAD design, reverse engineering, and small-scale manufacturing.

Feb 2024 – Oct 2025
1 year 9 months

- Maintained a 4.6-star client rating across completed engagements.
- Designed and iterated an accessibility device, including a tie-tying assistance tool.
- Reverse-engineered a broken replacement component from reference materials and technical drawings.
- Manufactured a low-volume production run of 30 tailored tablet holders.

Shape Machine, Social Media & Web Design Contributor

Atlanta, GA

Work with an early-stage startup developing shape-computation tools and supporting public-facing digital communication.

Sept 2026 – present
1 month

- Launched Shape Machine's digital presence through social-media outreach.
- Collaborated with an interdisciplinary team to publish the Shape Computation Lab website.

Wyzant, Freelance Private Tutor

Remote

Work directly with students of different ages to schedule lessons, prepare instructional materials, and tailor tutoring to each student's needs.

May 2026 – Aug 2026
4 months

- Tutored approximately 100 hours while maintaining a 5-star rating across approximately 70 reviews.
- Schedule lessons directly with students and prepare materials that clarify each lesson's subject matter.
- Developed online tutoring, time-management, lesson-planning, and student-communication skills.

Mathnasium of Dacula, Mathematics Instructor

Dacula, GA

Instructed students from foundational arithmetic through AP Calculus while adapting explanations to individual learning speeds.

Aug 2024 – Aug 2025
1 year 1 month

- Instructed up to five students simultaneously in a fast-paced learning environment.
- Translated complex mathematical concepts into clear, step-by-step explanations.
- Supported student confidence and academic progress across diverse grade levels.

Mercor & Handshake AI, AI Data Trainer & Labeler

Remote

Analyze AI-generated artifacts and author prompts to provide structured feedback on model quality.

May 2026 – present
5 months

Involvement

Technology Student Association, Treasurer (TSA)

Hoschton, GA

Managed club finances, supported engineering competition activities, and helped coordinate member communication.

Aug 2024 – May 2026

- Managed club finances and structural budgeting alongside faculty advisors.
- Organized community fundraising initiatives, including a local movie-night event.
- Organized and delivered presentations about upcoming events and member opportunities.
- Competed at the state level as part of a biotechnology team.

Languages

English

Native

German

Conversational

Interests

Advanced Manufacturing & Research: Additive Manufacturing, Rapid Prototyping, Robotics & Electromechanical Systems, Hardware & Electronics, Experimental Design, Applied Engineering Research

Systems, Software & AI: Programming, Operating Systems (Windows, macOS, Linux), Engineering Software, Local & Agentic AI Experimentation

Design, Craft & Personalization: Personalized Product Design, Everyday Problem Solving, Woodworking, Culinary Experimentation

Outdoors, Strategy & Learning: Hiking & Outdoor Exploration, Chess, Continuous Technical Learning